Predicate Logic – OSP-5 Notes

Definition

Predicate Logic, also known as **First-Order Logic (FOL)**, is a formal system in artificial intelligence used to represent facts and relationships **more precisely** than propositional logic. It allows the use of **variables**, **quantifiers**, and **predicates** to describe complex statements about objects and their properties.

Components of Predicate Logic

Component	Description
Predicate	A function that expresses a property or a relation. Example: `Loves(x, y)`
Terms	Can be constants, variables, or functions.
Quantifiers	Indicate the scope of a statement (see below).
Connectives	Logical operators such as AND (∧), OR (∨), NOT (¬), IMPLIES (→)

Types of Quantifiers

1. **Universal Quantifier (∀)**
    - Means "for all".

- Example: $\forall x\ Human(x) \rightarrow Mortal(x)$
  ("All humans are mortal")

2. **Existential Quantifier (∃)**
   - Means “there exists”.
   - Example: $\exists x\ Loves(x, Pizza)$
     ("There exists someone who loves pizza")

Example

Natural Language Statement:
"Every student loves AI."

Predicate Logic Representation:
$\forall x\ (Student(x) \rightarrow Loves(x, AI))$

Advantages of Predicate Logic

- Expresses **relationships** and **properties** clearly.
- Supports **inference and reasoning** using rules.
- More **powerful** than propositional logic.

Applications in AI

Area	Use Case

Expert Systems	Rule-based reasoning
Natural Language	Understanding and parsing sentences
Knowledge Bases	Storing facts and relationships
Robotics	Describing object properties and actions

Summary Table

Feature	Description
Level	First-order (allows variables)
Used For	Representing complex facts and relationships
Key Symbols	∀ (forall), ∃ (exists), →, ∧, ¬
Benefit	Enables intelligent reasoning and deduction